

Worked Answer Key

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1. $\frac{3}{8}$: 3 of 8 equal parts are shaded. 2. Numerator 5 counts parts; denominator 7 names equal parts in the whole. 3. Any bar split into 6 equal parts with 4 shaded; $\frac{4}{6}$.

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1. $\frac{3}{5}$ of 20 = $20 \div 5 \times 3 = 12$. 2. Blue $\frac{7}{12}$; not blue $\frac{5}{12}$. 3. $24 \div 4 = 6$ students.

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1. $\frac{1}{9}$. 2. Eight $\frac{1}{8}$ pieces make 1 whole. 3. $\frac{1}{3} > \frac{1}{6}$ because thirds are larger equal parts.

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1. $\frac{7}{4} = 1\frac{3}{4}$ (4 goes into 7 once, remainder 3). 2. $2\frac{3}{5} = (2 \times 5 + 3)/5 = \frac{13}{5}$. 3. One whole plus 1 of 2 parts; $1\frac{1}{2} = \frac{3}{2}$.

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1. Six equal intervals; fourth tick is $\frac{4}{6}$. 2. $\frac{3}{8}$. 3. $\frac{5}{4} = 1\frac{1}{4}$; mark one whole and one fourth.

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1. Multiply by 4: $\frac{2}{3} = \frac{8}{12}$. 2. Multiply by 3: $\frac{5}{8} = \frac{15}{24}$. 3. Examples: $\frac{6}{14}$ and $\frac{9}{21}$.

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1. Divide by 6: $\frac{12}{18} = \frac{2}{3}$. 2. Divide by 5: $\frac{20}{25} = \frac{4}{5}$. 3. Yes: $18 \div 6 = 3$ and $24 \div 6 = 4$, so $\frac{18}{24} = \frac{3}{4}$.

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1. Four fourths of each half make four eighths, so both cover one-half. 2. $\frac{3}{6} = \frac{1}{2}$; also $\frac{6}{12}$. 3. True: $\frac{2}{5} \times 2 = \frac{4}{10}$.

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1. Multiply by 5: $\frac{3}{4} = \frac{15}{20}$. 2. Multiply by 3: $\frac{7}{9} = \frac{21}{27}$. 3. Multiply by 3: $\frac{4}{11} = \frac{12}{33}$.

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1. Divide by 2: $\frac{6}{10} = \frac{3}{5}$. 2. Divide by 4: $\frac{8}{12} = \frac{2}{3}$. 3. Divide by 5: $\frac{15}{35} = \frac{3}{7}$.

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1. Divide by 4: $\frac{16}{20} = \frac{4}{5}$. 2. Divide by 7: $\frac{21}{28} = \frac{3}{4}$. 3. Yes; 9 and 13 have no common factor greater than 1.

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1. Yes: $\frac{2}{6}$ divides by 2 to give $\frac{1}{3}$. 2. Multiply by 2: $\frac{3}{4} = \frac{6}{8}$ cup. 3. Divide by 2: $\frac{6}{10} = \frac{3}{5}$ mile.

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1. $\frac{2}{3} = \frac{8}{12}$, so =. 2. $\frac{5}{6} = \frac{15}{18}$, so =. 3. $\frac{3}{5} = \frac{9}{15}$, so =.

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1. Split one rectangle into thirds with 2 shaded; split another into ninths with 6 shaded. Both show $\frac{2}{3} = \frac{6}{9}$. 2. Divide into fourths with 3 shaded, or twelfths with 9 shaded. 3. Five groups with 2 shaded each gives $\frac{2}{5}$; fifteen parts with 6 shaded gives $\frac{6}{15}$.

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1. Multiplying by 3 makes three copies of every original part: $a/b = (3a)/(3b)$, so the amount stays the same. 2. The denominator also changes; $4/8$ and $1/2$ are equal, not greater. 3. Multiply by 4: $5/6 = 20/24$.

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1. $9/4 = 2\frac{1}{4}$; converting back gives $(2 \times 4 + 1)/4 = 9/4$. 2. $1\frac{2}{3} = (1 \times 3 + 2)/3 = 5/3$. 3. Yes: $2\frac{1}{2} = (2 \times 2 + 1)/2 = 5/2$.

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1. If 4 is $2/3$, one third is 2, so the whole is $2 \times 3 = 6$ parts. 2. $6 \div 3 = 2$ per fifth; $2 \times 5 = 10$ tiles. 3. $15 \div 5 = 3$ per eighth; $3 \times 8 = 24$ cm.

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1. Invalid: adding changes the value. Multiply both terms by 2: $3/5 = 6/10$. 2. Not simplest: $6/8$ divides by 2 to $3/4$. 3. The denominator was multiplied by 2 without multiplying the numerator; correct form is $2/7 = 4/14$.

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1. $24/32$ divides by 8 to $3/4$; denominator 20 requires multiply by 5: $15/20$. 2. $18/30$ divides by 6 to $3/5$; denominator 5 fraction is $3/5$. 3. $3/4 < 4/4 < 3/2$ (0.75, 1, 1.5).

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1. $18/24$ divides by 6 to $3/4$; multiply by 2 for denominator 8: $6/8$. 2. Examples: $8/10$ ($\times 2$), $12/15$ ($\times 3$), $20/25$ ($\times 5$). 3. Multiply numerator and denominator by the same nonzero whole number; $7/10 \times 4 = 28/40$.